Peer Review Report

Manuscript: "TEMPC: UAV + UGV" — Event-Triggered Tube MPC for Bidirectional Observability-Aware Aerial–Ground Cooperation

Venue	ICC 2026 — 12th Indian Control Conference (ICC-12), IEEE Xplore–indexed
Review type	Full technical review (concept-level)
Recommendation	Reject — major revision required before the manuscript is reviewable as a paper

Bottom line. The core idea is genuinely interesting and well motivated, but the submission is in an unfinished draft state, contains **no results whatsoever**, and has several conceptual gaps in the very theory it claims as its contribution. The review below separates the *idea* (promising) from the *execution* (not yet there).

Summary of claims

The paper proposes a coupled aerial–ground tracking system: a quadrotor UAV observes a unicycle UGV via a downward camera and an ArUco fiducial marker. Three contributions are claimed: (1) a belief-space NMPC for the UGV that penalizes predicted estimation covariance and field-of-view (FoV) exit, so the *target* actively keeps itself observable; (2) an adaptive tube-MPC for the UAV whose tube radius scales with the EKF covariance; and (3) a perception-driven event-trigger that replans only on covariance spikes, anticipated path curvature, or tube violations.

What is good (keep this)

The framing — that passive "minimize distance to target" trackers fail under occlusion and aggressive maneuvers — is correct and well argued. The *bidirectional* coupling, in which the target shapes its own trajectory to stay observable while the tracker adapts its robustness to perception quality, is a legitimately novel angle; most active-perception work makes the sensor smart and the target dumb. The view-quality-dependent measurement covariance $R_k(q_k)$ is a sensible modeling choice. And the mRPI derivation (Eqs. 7–10), following Raković et al. 2005, is done correctly: the truncated Minkowski series scaled by $1/(1 - \rho^{N+1})$ is the standard, valid outer approximation. The bones are there.

Major technical concerns

1. The adaptive tube breaks the invariance guarantee it relies on, and the feasibility argument is backwards.

This is the most serious conceptual problem. Tube MPC works because Ω is a *robust positively invariant* (RPI) set satisfying $A_K\Omega \oplus W \subseteq \Omega$, computed offline for a fixed, bounded disturbance set, after which constraints are tightened once via $X = X \ominus \Omega$. Equation (6) then scales the tube online by $\alpha_k = \max(1, \beta \cdot \mathrm{tr}(\Sigma_{xy}))$. Two problems: (a) the offline invariance condition no longer holds for $\alpha_k\Omega$ unless it is re-derived — a scaled RPI set is not automatically RPI; and (b) the paper claims that expanding α_k "prevents infeasibility," but a *larger* tube means a *larger* Pontryagin difference $X \ominus \alpha_k\Omega$, i.e. *tighter* nominal constraints and a *smaller* feasible region — the opposite of what is claimed. There is mature theory for time-varying tubes (homothetic / parameterized tube MPC, Raković et al.) that handles exactly this, but it requires the scaling to be part of the optimization with its own recursive-feasibility argument. As written, the central contribution has no guarantee.

2. Bounded-disturbance tube MPC is grafted onto an unbounded stochastic noise model with no reconciliation.

The effective disturbance is written $w_{eff} = w_a - BKv_k$ (measurement noise entering through feedback), and the tube/mRPI machinery requires a *bounded* set W. But v_k and $w_g \sim N(0, Q_g)$ are Gaussian with unbounded support, and the entire UGV side is an EKF tracking a covariance Σ. One cannot form a hard RPI set from Gaussian noise without truncating to a confidence region — and that truncation, its probability level, and how it links Σ to the disturbance bound W are never stated. The paper freely mixes the stochastic framework (Σ, EKF, covariance) with the set-bounded one (Ω, mRPI, hard constraints). Pick one and be rigorous, or explicitly bridge them (e.g. probabilistic reachable sets / stochastic tube MPC).

3. No analysis of the coupled closed loop — which is the actual contribution.

The stability/feasibility apparatus (DARE terminal cost, terminal invariant set X_f) applies only to the UAV's *nominal tracking* subproblem in isolation. But the headline is the *bidirectional* coupling: the UGV's cost depends on $\hat{x}_a$ and on predicted Σ, while the UGV's Σ depends on the UAV's future positions through $R_k(q_k)$. Two MPCs each optimizing against a prediction of the other can chatter, oscillate, or fail to reach a consistent equilibrium. There is no analysis (or even discussion) of the coupled system's stability, of the information structure (does the UGV know the UAV's planned trajectory? assumed? communicated? at what rate?), or of whether a fixed point exists. Without this, "bidirectional coupling" is an architecture, not a result.

4. The belief-space NMPC undercuts the paper's own computational thesis.

The selling point is *reducing* computation via event-triggering on the UAV's linear tube-MPC. But the UGV side adds a belief-space NMPC (Eq. 4) that must propagate the EKF covariance through the horizon to penalize $tr(\Sigma_k)$ — non-convex and expensive — with no event-triggering, plus a view-quality-maximizing nominal planner. The computational accounting is entirely one-sided. The abstract's claim of "reduced computational load *but less error*" (a strict improvement on both axes at once) is a strong claim that event-triggering normally cannot deliver: the usual story is large compute savings for a small performance cost. This must be measured, not asserted.

5. "Lemma 1" is a mechanism, not a proven lemma.

The fractional trigger $\max(H_\Omega e_k - \sigma h_\Omega) > 0$ and the contracted set Ω_σ are intuitive, but Lemma 1 has no proof. The claim that "while $e \in \Omega_\sigma$ the shifted previous sequence is retained" needs a guarantee that the ancillary controller keeps $e \in \Omega$ between triggers and that recursive feasibility survives the skipped solves — precisely what an event-triggered MPC lemma must establish. Also unspecified: how σ (the contraction) interacts geometrically with α_k (the expansion) when both act on Ω at once. State it as a proposition with a real proof, or call it a heuristic.

6. The view-quality metric is non-monotone in the wrong direction at altitude.

In Eq. (2), as $z_a \to \infty$ for a centered target, $q_k \to 1$ — i.e. infinite altitude gives perfect view quality. Physically the opposite happens: the ground sampling distance grows, the ArUco marker falls below detectable pixel resolution, and the marker is lost. The model captures lateral FoV and a low-altitude floor but is missing the resolution / GSD penalty that should make q degrade at *high* altitude. As written, the planner has an incentive to climb without bound.

Completeness / venue-fit issues (these alone would force a desk reject)

No experiments, no results, no figures. The abstract says the method was "tested under extensive and varying adversarial conditions" and the conclusion says it was "experimentally demonstrated," yet there is not a single plot, number, simulation setup, or baseline comparison. The only table is the template placeholder (One / Two / Three / Four) and the only figure is the IEEE template's magnetic-core inductance plot. For a venue that explicitly prizes experimental validation, this is fatal in the current form. At minimum a simulation study is needed with an

ablation (passive vs. active UGV; fixed vs. adaptive tube; continuous vs. event-triggered) and the compute-vs-tracking-error tradeoff curve over σ.

The manuscript still contains the LaTeX template scaffolding. Section II ("Related Work") is literally the template line "A. Selecting a Template (Heading 2)"; Section III ("Problem Formulation") is empty; Section VI ("Using the Template") and the Fig. 1 caption are boilerplate; author names and affiliations are placeholders ("Albert Author," "Bernard D. Researcher," papercept.net); and multiple "[TODO: Cite …]" markers remain.

Empty related work means no positioning against prior art. Every ingredient exists in the literature (Mayne et al. tube MPC; Raković mRPI and homothetic tubes; event-triggered MPC, e.g. Eqtami et al.; perception-aware planning, e.g. Falanga et al. PAMPC and Zhang–Scaramuzza; belief-space planning, Platt / van den Berg; cooperative UAV–UGV tracking). The TODOs show the authors know this; the novelty claim is unverifiable until the bidirectional coupling and perception trigger are situated against these.

Minor / specific comments

- Abstract grammar and over-claiming: "a event triggered," "adversial," "less error" (less than *what*?). Tighten and quantify.
- Section numbering is broken throughout: double labels ("A. B. mRPI Computation," "B. C. Fractional Event-Triggering"); the event-trigger cites "Section III.B" for the nominal planner, which actually lives in V.B (III is empty); an equation reference appears as a literal "(??)".
- Eqs. (18)–(19) are typographically garbled (stray asterisks/pipes, "*0—ti ≤ h∗Ω"). The terminal cost names P from the DARE but the displayed term appears to use x_{ref}, not P. Re-typeset and verify.
- Model inconsistency: the body calls the UAV a "quadrotor" and uses a 6-state double integrator (point mass), while the conclusion calls it "a simplified double integrator model." A quadrotor is underactuated and nonlinear; either justify the point-mass abstraction explicitly (differential flatness, inner-loop attitude controller) up front, or drop "quadrotor."
- $R_{max}(1 - q_k)$ in Eq. (3): clarify that R_{max} is a matrix and that dimensions are consistent.
- No values or tuning guidance for c_1, c_2, β, σ, λ, τ_{cov}, $\tau_к$; no algorithm box; no system diagram.
- "Temporal Drift" (max time since replan) makes this a hybrid event-/self-triggered scheme — fine, but name it precisely.

Path to a strong submission

The idea deserves to become a paper. Concretely: (1) commit to one uncertainty framework and re-derive the adaptive tube as homothetic / parameterized tube MPC with a real recursive-feasibility result; (2) state and *prove* the event-trigger lemma, including its interaction with the adaptive radius; (3) say something rigorous about the coupled UGV–UAV loop (information structure plus at least a stability or non-chattering argument); (4) fix the view-quality model's altitude behavior; and above all (5) run the simulation study with ablations and the σ tradeoff curve, ideally with a hardware demonstration given ICC's experimental emphasis. Then write the related work and remove all template residue.
